

Solution

By comparing with spectral charts in appendix 3, we notice that

- H-lines are the most prominent lines,
- He I feature at $4144 \text{ \AA}$ is barely visible and
- Ca I feature around $4230 \text{ \AA}$ is not seen.

Thus, it may be reasonable to restrict the range of spectral types to B5-A5.

We find that the ratio of the intensity of a pair of lines and compare it with the standard spectra. We see, $\text{Ca II}/\text{He I} \sim \frac{1}{4}$

For B8 standard spectra it is $\sim \frac{1}{5}$, for A0 standard spectra it is $\sim \frac{1}{3}$.